

Coil Sanity Check

Theory and hand calculation, verified against FEMM

Coil Sanity Check — Theory and Hand Calculation

A step-by-step hand calculation of the axle-counter coil: turns, inductance, magnetic field, flux, resistance, resonance, and mutual inductance.

Each step states **the theory first**, then applies it to the actual coil. Every number is checked against the FEMM simulation at the end.

Prepared by Rudranarayan.

0. The coil we are calculating

Every value below comes from [axlecounter_3dphysics/config.py](#), which is the single source of truth for the whole project.

Symbol	Quantity	Value
r_i	Inner winding radius	30 mm
r_o	Outer winding radius	40 mm
R	Mean radius $(r_i+r_o)/2$	35 mm
l	Winding length (axial)	25 mm
c	Winding depth (r_o-r_i)	10 mm
—	Wire	18 AWG, diameter 1.024 mm
a	Wire radius	0.512 mm
—	Packing fraction	0.70
f	Frequency	20 kHz
I	Drive current (peak)	2.5 A
d	Coil-to-coil separation	164.06 mm

Constants: $\mu_0 = 4\pi \times 10^{-7}$ H/m, $\rho(\text{copper}) = 1.68 \times 10^{-8}$ $\Omega \cdot \text{m}$.

Shape matters. The coil is 70 mm across but only 25 mm long, so $l/R = 0.71$. It is a **short, fat** coil — not a long thin solenoid. Step 2 shows this is the single most important fact in the whole calculation.

1. How many turns fit?

Theory

A coil is wound into a rectangular **winding window** of depth c and length l . Each turn of wire occupies a circle of area πa^2 . Real wire cannot fill the window perfectly — there are gaps between round wires and a little insulation — so we multiply by a **packing fraction**, typically 0.65–0.75 for hand-wound round wire.

$$N = \frac{\text{packing} \times (\text{window area})}{\text{area of one wire}} = \frac{p \cdot c \cdot l}{\pi a^2}$$

Calculation

Window area:

$$c \times l = 0.010 \times 0.025 = 2.500 \times 10^{-4} \text{ m}^2$$

Area of one 18 AWG wire:

$$\pi a^2 = \pi (0.512 \times 10^{-3})^2 = 8.2354 \times 10^{-7} \text{ m}^2$$

So:

$$N = \frac{0.70 \times 2.500 \times 10^{-4}}{8.2354 \times 10^{-7}} = 212.49$$

N = 212 turns (rounded to a whole turn).

Rounding 212.49 down to 212 makes the real copper fill 0.6984 instead of 0.700. That exact figure is written into the FEMM material as its fill factor, so the simulation models the same winding density we assumed here.

2. Self-inductance L

Theory

Inductance measures the flux a coil links per amp of its own current. There are three standard formulas, and **choosing the right one is the whole problem**:

(a) Long solenoid — assumes the coil is much longer than it is wide:

$$L = \frac{\mu_0 N^2 \pi R^2}{l} \quad \text{needs } l \gg R$$

(b) Single loop — assumes all N turns sit in one thin filament:

$$L = \mu_0 N^2 R \left(\ln \frac{8R}{a} - 2 \right) \quad \text{needs } a \ll R$$

(c) Wheeler's multilayer formula — an empirical fit built for exactly our case: a coil with real length and real winding depth. In inches it reads $L[\mu\text{H}] = 0.8a^2N^2/(6a+9b+10c)$. Converting to metres, every length divides by 0.0254; the numerator has two lengths squared and the denominator one, so one factor of 1/0.0254 survives:

$$L = \frac{0.8}{0.0254} \times 10^{-6} \frac{R^2 N^2}{6R + 9l + 10c} = \frac{31.496 \times 10^{-6} R^2 N^2}{6R + 9l + 10c}$$

Which one applies?

Formula	Requires	Our coil	Verdict
Long solenoid	$l \gg R$	$l/R = \mathbf{0.71}$	fails
Single loop	thin bundle	bundle is 10×25 mm vs R = 35 mm	fails
Wheeler	winding length > 0.2R	0.71 R	use this

Calculation

Denominator:

$$6(0.035) + 9(0.025) + 10(0.010) = 0.210 + 0.225 + 0.100 = 0.535$$

Numerator:

$$31.496 \times 10^{-6} \times (0.035)^2 \times (212)^2 = 31.496 \times 10^{-6} \times 1.225 \times 10^{-3} \times 44944 = 1.7341 \times 10^{-3}$$

$$L = \frac{1.7341 \times 10^{-3}}{0.535} = 3.2412 \times 10^{-3} \text{ H}$$

L = 3.241 mH

Proof that the other two are wrong

$$L_{\text{solenoid}} = 8.694 \text{ mH} \quad L_{\text{loop}} = 8.508 \text{ mH}$$

That is **2.68x** and **2.63x** too big respectively.

FEMM solved this same coil axisymmetrically and got 3.180 mH. Wheeler's 3.241 mH is **1.9 % away** — well inside its ~2 % accuracy. The other two formulas are off by a factor of 2.6. This is the calculation validating itself.

3. Magnetic field at the centre

Theory

For an **infinitely long** solenoid with n turns per metre, the field inside is uniform:

$$B = \mu_0 n I$$

A real coil of finite length leaks flux out of its ends, so the true centre field is always **lower**. For a coil of length l and radius R :

$$B_{\text{centre}} = \mu_0 n I \cdot \frac{l/2}{\sqrt{R^2 + (l/2)^2}}$$

The fraction on the right approaches 1 only when $l \gg R$.

Calculation

Turn density:

$$n = N/l = 212/0.025 = 8480 \text{ turns/m}$$

Infinite-solenoid value:

$$\mu_0 n I = (1.2566 \times 10^{-6})(8480)(2.5) = 26.64 \text{ mT}$$

Finite-length correction factor:

$$\frac{0.0125}{\sqrt{0.035^2 + 0.0125^2}} = \frac{0.0125}{0.037165} = 0.3363$$

$$B_{\text{centre}} = 26.64 \times 0.3363 = 8.960 \text{ mT}$$

B = 8.96 mT

The real coil produces only **34 %** of the infinite-solenoid field — the same short-and-fat geometry that disqualified the solenoid formula in Step 2.

4. Magnetic flux and flux linkage

Theory

Flux Φ is the field passing through **one** turn. **Flux linkage** λ counts all N turns:

$$\Phi = B \cdot A, \quad \lambda = N\Phi, \quad \lambda = LI$$

Keeping these apart is the most common source of factor-of-N errors. FEMM reports **linkage**, not flux.

Calculation

Mean turn area:

$$A = \pi R^2 = \pi(0.035)^2 = 3.8485 \times 10^{-3} \text{ m}^2$$

From the inductance:

$$\lambda = LI = 3.2412 \times 10^{-3} \times 2.5 = 8.103 \times 10^{-3} \text{ Wb} \cdot \text{turns}$$

$$\Phi = \lambda/N = 8.103 \times 10^{-3}/212 = 38.22 \text{ } \mu\text{Wb}$$

$\Phi = 38.22 \text{ } \mu\text{Wb}$ per turn, $\lambda = 8.10 \text{ mWb} \cdot \text{turns}$

Cross-check using the field from Step 3

$$\Phi \approx B_{\text{centre}} \cdot A = 8.960 \times 10^{-3} \times 3.8485 \times 10^{-3} = 34.48 \text{ } \mu\text{Wb}$$

The two routes differ by **11 %**, which is the expected accuracy when a 10×25 mm multilayer winding is represented by one mean radius and one centre-field value. Agreement at this level confirms no gross error.

5. Resistance and skin effect

Theory

At DC, resistance is just the wire's length over its cross-section:

$$R_{DC} = \frac{\rho l_{\text{wire}}}{\pi a^2}$$

At AC, current is pushed toward the conductor surface. The characteristic depth is the **skin depth**:

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$

If the wire radius exceeds δ , the usable copper shrinks and resistance rises. With $x = a/\delta$ the standard approximation has two branches.

For a thin wire ($x < 1$), the correction is tiny:

$$\frac{R_{AC}}{R_{DC}} = 1 + \frac{x^4}{48}$$

For a thick wire ($x \geq 1$), resistance grows roughly in proportion to x :

$$\frac{R_{AC}}{R_{DC}} = \frac{x}{2} + \frac{3}{4} + \frac{3}{32x}$$

Calculation

Wire length:

$$l_{\text{wire}} = N \cdot 2\pi R = 212 \times 2\pi \times 0.035 = 46.62 \text{ m}$$

$$R_{DC} = \frac{1.68 \times 10^{-8} \times 46.62}{8.2354 \times 10^{-7}} = 0.951 \Omega$$

Skin depth at 20 kHz:

$$\delta = \sqrt{\frac{1.68 \times 10^{-8}}{\pi(20000)(4\pi \times 10^{-7})}} = 0.4613 \text{ mm}$$

Since $x = 0.512/0.4613 = \mathbf{1.110} \geq \mathbf{1}$, use the second branch:

$$\frac{R_{AC}}{R_{DC}} = \frac{1.110}{2} + 0.75 + \frac{3}{32(1.110)} = 0.555 + 0.750 + 0.084 = 1.389$$

$$R_{AC} = 0.951 \times 1.389 = 1.321 \Omega$$

R_AC = 1.32 Ω (39 % above DC)

δ is smaller than the wire radius, so skin effect is real here and cannot be ignored. A thinner wire, or litz wire, would reduce it.

6. Quality factor and resonant tuning

Theory

Q compares stored energy to energy lost per cycle:

$$Q = \frac{\omega L}{R_{AC}}$$

To cancel the coil's reactance we add a capacitor that resonates with it:

$$\omega = \frac{1}{\sqrt{LC}} \implies C = \frac{1}{\omega^2 L}$$

At resonance the supply only has to overcome R_{AC} — but the **reactive voltage does not disappear**, it appears across the capacitor:

$$V_{cap} = I \omega L$$

Calculation

$$\omega = 2\pi(20000) = 1.2566 \times 10^5 \text{ rad/s.}$$

$$Q = \frac{1.2566 \times 10^5 \times 3.2412 \times 10^{-3}}{1.321} = 308$$

$$C = \frac{1}{(1.2566 \times 10^5)^2 \times 3.2412 \times 10^{-3}} = 19.54 \text{ nF}$$

$$V_{cap} = 2.5 \times 1.2566 \times 10^5 \times 3.2412 \times 10^{-3} = 1018 \text{ V}$$

Q = 308, C = 19.5 nF, V_cap = 1018 V peak

Design warning. Resonance does not remove the high voltage — it *moves* it from the supply onto the capacitor. Driving 2.5 A through this coil needs a capacitor rated **above 1 kV**. This is why the design scripts size parts against 250 / 630 / 1000 / 2000 V capacitor classes.

7. Mutual inductance M

Theory

Two coils sit side by side, both axes vertical, separated horizontally by d.

Far from a coil, its field looks like a **magnetic dipole** of moment:

$$m = NIA$$

A dipole's field depends on direction. Directly above it (on-axis) the field is twice as strong as it is out to the side (equatorial):

$$B_{axis} = \frac{\mu_0 2m}{4\pi d^3}, \quad B_{equatorial} = \frac{\mu_0 m}{4\pi d^3}$$

Because both axes are vertical and the coils are side by side, **each sits in the other's equatorial plane**, so we use the second one. Assuming the field is uniform across the secondary and equal to its value at the secondary's centre:

$$\Phi_{1 \rightarrow 2} = B_{eq}A, \quad \lambda_2 = N\Phi_{1 \rightarrow 2}, \quad M = \frac{\lambda_2}{I}$$

Combining gives the closed form:

$$M = \frac{\mu_0 \pi N^2 R^4}{4d^3}$$

Calculation

Dipole moment:

$$m = 212 \times 2.5 \times 3.8485 \times 10^{-3} = 2.040 \text{ A}\cdot\text{m}^2$$

Equatorial field at the far coil (d = 0.16406 m):

$$B_{eq} = \frac{(4\pi \times 10^{-7})(2.040)}{4\pi(0.16406)^3} = \frac{10^{-7} \times 2.040}{4.4157 \times 10^{-3}} = 46.19 \text{ }\mu\text{T}$$

Flux through one secondary turn:

$$\Phi = 46.19 \times 10^{-6} \times 3.8485 \times 10^{-3} = 177.8 \text{ nWb}$$

Linkage and M:

$$\lambda_2 = 212 \times 177.8 \times 10^{-9} = 37.69 \text{ }\mu\text{Wb}\cdot\text{t}$$

$$M = \frac{37.69 \times 10^{-6}}{2.5} = 15.07 \text{ }\mu\text{H}$$

M = 15.07 μH (hand estimate)

FEMM measured M = 21.47 μH — the simulation is **1.42x higher**.

Why the hand value is low, and why that is expected. The dipole formula pretends each coil is a *point* at its centre, 164 mm away. The real coils are 70 mm wide, so their facing conductor bundles are only about 94 mm apart — much closer than 164 mm — and closer conductors couple far more strongly (M falls as 1/d³). Working the other way, the steel rail between the coils shields some flux. The point-dipole result is therefore an **order-of-magnitude estimate**, good for showing how M *scales*, not for predicting its exact value. FEMM, which models the true shape and the rail, is the authority.

8. Coupling coefficient and output signal

Theory

The coupling coefficient states what fraction of one coil's flux reaches the other:

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{M}{L} \quad (\text{identical coils})$$

The induced open-circuit signal follows from Faraday's law in AC form:

$$V_{rx} = \omega M I$$

Calculation

$$k = \frac{21.47 \times 10^{-6}}{3.2412 \times 10^{-3}} = 0.0066$$

$$V_{rx} = 1.2566 \times 10^5 \times 21.47 \times 10^{-6} \times 2.5 = 6.74 \text{ V}$$

k = 0.66 %, V_{rx} = 6.74 V

Very weak coupling — and that is correct for an axle counter. The coils are deliberately far apart. The signal is not the coupling itself but the **change** in it: when a steel wheel enters the gap, FEMM shows M collapsing from 21.47 μH to 2.58 μH , an **88 % dip**, with RX voltage falling 6.74 V $\rightarrow$ 0.81 V. That dip is the axle count.

9. Hand calculation vs simulation

Quantity	Hand calculation	FEMM	Agreement
Turns N	212.49 $\rightarrow$ 212	212	exact
Inductance L	3.241 mH	3.180 mH	1.9 %
L (solenoid formula)	8.694 mH	—	2.68 $\times$ — formula invalid
L (single-loop formula)	8.508 mH	—	2.63 $\times$ — formula invalid
Wire length	46.62 m	46.6 m	exact
R_DC	0.951 Ω	0.951 Ω	exact
R_AC	1.321 Ω	1.321 Ω	exact
Q	308	309	exact
C resonant	19.54 nF	19.47 nF	0.4 %
V_cap	1018 V	1022 V	0.4 %
Mutual inductance M	15.07 μH	21.47 μH	1.42 $\times$ — see Step 7
V_rx	4.74 V (from hand M)	6.74 V	1.42 $\times$

Conclusion. Everything that depends on the coil *by itself* — turns, inductance, field, flux, resistance, Q, resonance — matches the simulation to within 2 %, most of it exactly. That validates the solver setup and the geometry.

The only quantity that disagrees is the coil-to-coil mutual inductance, and it disagrees for a known and quantified reason: the point-dipole approximation cannot represent two 70 mm-wide coils only 164 mm apart.

10. Assumptions, and how much they matter

Assumption	Status	Effect
Packing fraction 0.70	design choice; sets N	direct, $N \propto p$
Wheeler valid (winding length > 0.2R)	satisfied	1-2 %
Turns rounded 212.49 → 212	true fill 0.6984	0.2 %
Coil is one filament at R = 35 mm	winding is really 10×25 mm	~10 %
Uniform field across the far coil	weak — see below	large
Point dipole ($d \gg R$)	weak — $d/R = 4.7$	1.42×
No steel rail in the hand model	rail shields flux	opposes the above

The uniform-field assumption, quantified. Field falls as $1/d^3$, so across the far coil's own width (± 35 mm about 164 mm):

$$\frac{B(d-R)}{B(d)} = \left(\frac{164.06}{129.06}\right)^3 = 2.05, \quad \frac{B(d+R)}{B(d)} = \left(\frac{164.06}{199.06}\right)^3 = 0.56$$

The true field varies from **2.05×** down to **0.56×** the centre value. Calling it uniform is a deliberate simplification, not an exact result.

Rule of thumb: trust the hand calculation for **scaling laws** — $L \propto N^2$, $M \propto N^2$, $M \propto 1/d^3$, $M \propto R^4$. Trust FEMM for **absolute** mutual inductance, because only it models the real coil shape and the steel rail.

Appendix — reproducing these numbers

```
cd axlecounter_3dphysics
py -3 analysis_and_reporting/sanity_check.py
```

That script recomputes every value above from [config.py](#), solves the same coil in FEMM, and writes [reports/sanity_check_report.md](#). Current status: **28 checks passed, 0 failed**.

To re-run the entire study (all FEMM solves, analytics, notebooks, figures):

```
cd axlecounter_3dphysics
py -3 run_all.py
```